Deutsche Bank
Research

Rating
Sell

North America
United States

Health Care
Medical Supplies & Devices

Company
Intuitive Surgical

Reuters	Bloomberg	Exchange	Ticker
ISRG.OQ	ISRG US	NMS	ISRG

Date
23 July 2026

Industry Update

Price at 22 Jul 2026 (USD)	340.69
Price Target	324.00
52-week range	592.85 - 340.69

Key takeaways from Society of Robotic Surgery investor summit

Valuation & Risks

Imron Zafar
Research Analyst
+1-212-250-3676

Pito Chickering
Research Analyst
+1-917-635-9954

Kieran Ryan
Research Analyst
+1-212-250-6879

Ben Shaver
Research Associate
+1-212-250-9926

Ahead of the SRS scientific sessions that kick off Thursday, today the Society held its annual investor symposium featuring program updates from senior executives of several robotic surgery companies as well as KOL panel discussions spanning an array of hot topics du jour including technology innovation, site of care, AI, and remanufactured instruments. This morning's earlier-than-expected US FDA approval of J&J's Ottava generated considerable buzz, and herein we present key takeaways from panel discussions with leadership of J&J, Intuitive, Medtronic, and Stryker.

We also highlight some interesting findings from a large survey of SRS members (n=451 surgeons across specialties from around the world) that help inform the outlook for robot-assisted surgeries: technology adoption/procedure growth, competitive landscape, and structural trends like continued surgical volume migration to the ASC and expected impact of AI.

SRS physician survey

- On average, robot-assisted surgery penetration is expected to increase from 33% presently to 48% five years hence on a worldwide basis. This implies a WW robotic surgery procedure CAGR of +~11% over this timeframe and continuing to be driven both by conversion of open to a MIS approach broadly and, within the latter, further shift from conventional lap to robot-assisted surgery.
- In looking at the US survey respondent subgroup, robotic penetration is projected to rise from 64% currently to 69% over this timeframe – thus implying more modest robotic surgery volume growth relative to OUS/WW rates, which of course reflects a more penetrated US market vs internationally. This expected increase in US robotic penetration is expected to come exclusively from conversion of open surgeries as conventional lap usage is projected to remain flattish.
- Notably, given the selection bias of this survey targeting SRS membership (with a median of 17 years of experience as an attending), technology adoption amongst this cohort is obviously higher than the overall market and hence this growth rate surely understates the overall market outlook. Indeed, the consensus view of all the companies remains that the overall soft-tissue robotic surgery global TAM is still <10% penetrated.
- Regarding competitive landscape, on a worldwide basis, Intuitive's da Vinci currently holds an 81% installed base share, which is forecasted to

decline to 68% over the next five years, at which point Medtronic is expected to be the #2 player with Hugo garnering 10% terminal share amongst survey respondents.

- In the US, da Vinci's installed base share currently stands at 98% and likewise is expected to decline in the years ahead – though expectations for Intuitive to maintain a still-dominant 84% share five years hence is notable in light of market entry of several competitors including Medtronic and J&J, who are entering this market with the advantage of a longstanding strong presence in surgical devices. One caveat here is that the survey was conducted in March, and given the rapid developments in this category (case in point being today's FDA approval of Ottava), share dynamics will continue to be fluid – but the clear takeaway is that da Vinci remains well-positioned for clear and sustained category leadership.
- Regarding site of care, survey data project that roughly one-third of all addressable surgeries will be done in the ASC setting five years hence. For frame of reference, we believe Intuitive's mix of procedures and installed base currently stands in the ~MSD range.
- In terms of AI, survey data clearly portend a significant impact on various aspects of robotic surgery, with intra-operative anatomic intelligence and pre-operative planning expected to be among the most important benefits at least initially.

J&J Ottava

- While J&J's receipt of FDA approval for Ottava was widely expected, this morning's timing was ahead of expectations (as evidenced by the fact that J&J did not come to SRS prepared to showcase the system to surgeon attendees).
- Management pointed out that the de novo pathway to FDA approval (vs 510k) is noteworthy in terms of demonstrating the technology differentiation and simply not having a predicate necessary for the latter pathway.
- One key differentiator of Ottava that will clearly be central to the "sales pitch" is the significantly smaller footprint, taking up 30-50% less space in the OR – which will surely be a competitive advantage particularly in the ASC channel.
- Initial FDA clearance includes a handful of general surgery procedures, with management noting that a steady cadence of additional indication approvals across other key specialties like urology and gynecology (and additional general surgery approvals like hernia) should follow in fairly short order.
- Along with these additional indication approvals, J&J's pipeline will yield a steady output of new offerings ranging from digital, instrumentation, software, and beyond.
- In terms of commercial strategy, like Intuitive, J&J plans to offer flexible Ottava acquisition models, ranging from outright capex purchases to operating leases to per-click installations.
- Notably, unlike Medtronic (and the typical medtech playbook in general), J&J made a deliberate strategic decision to pursue US FDA approval first rather than initially targeting OUS markets with an easier/faster pathway to regulatory clearance – which management cited as a sign of confidence in the platform.
- In terms of OUS, management is next prioritizing the Japan and Western European markets, though refrained from providing estimated timelines

here.

- While the company acknowledges the market dominance of da Vinci as it enters the soft tissue robotic surgery market decades later than Intuitive, confidence remains high that Ottava represents a major growth vector for JNJ longer term given Ethicon's strong incumbent presence in MIS and the fact that global TAM penetration is still single-digits – hence the focus should remain more on market growth vs market share.

Intuitive da Vinci

- To preface, this panel discussion was pre-recorded a couple days prior to last week's 2Q results and hence the discussion did not hit on the sequential US volume slowdown nor management's reaction to this morning's approval of Ottava.
- Management is increasingly focused on the ASC channel given the continued shift of lower acuity soft tissue surgeries to this site of care. While the current da Vinci business mix (both installed base and procedures) in the ASC segment is currently single digits, the company noted that ~70% of all US ASCs are aligned with IDNs that already have da Vinci systems in the hospital setting – and as such is working collaboratively with these customers as they reallocate the lower acuity cases to the surgery center and thus freeing up more capacity for more complex surgeries to be done in the acute hospital. As more hospitals upgrade to dV5, uptake of XiR in the ASC channel has been quite good given XiR's lower price point and the much greater cost sensitivity of ASCs.
- Launch of EUP round two in 2027 is expected to yield higher technology adoption by lowering the per-procedure I&A expense specifically for high-volume, lower acuity surgeries, though we strongly suspect this was in part also a defensive strategic move given the intensifying competition from other robotic platforms as well as remanufactured instruments.
- On remanufactured instruments, the company again noted that it has thus far not seen any meaningful hit from remanufactured instrument adoption and remains largely dismissive of this threat going forward, positing that the value proposition extends beyond just instrument ASP to include reliability, safety, etc. Our checks continue to indicate a much greater risk to the I&A segment relative to the company's stance.
- As discussed below, Stryker's CEO again noted today that they are watching this remanufactured da Vinci instrument space with high interest.
- As evidenced by last week's disclosure that a GI robot is in late-stage development, Intuitive has increased its investments in new platforms in pursuit of adjacent opportunities – with the "quintuple aim" remaining foundational to the investment strategy. No additional detail was disclosed on the GI platform relative to timelines etc.
- In terms of capital deployment, organic investment remains the top priority (particularly on R&D), though external opportunities will continue to be evaluated in a focused and disciplined manner.

Stryker

- CEO Kevin Lobo, whose then-controversial venture into Mako in 2013 has, and continues to, pay off in spades as adoption of robotic-assisted recon surgery continues to increase and expand into new categories.
- While Mako now accounts for 70% of US and 50% of Stryker's global knee recon procedures, confidence is high that there still remains ample runway for continued growth/share gains in both the US and especially

internationally. Ditto for Mako hip where penetration is still considerably lower and where uptake of revisions is seeing increasing momentum. Mako revision knee remains a work in progress, and while confidence here is high, no timelines were provided.

- With Stryker's OUS sales mix remaining well below large cap medtech peers, this remains a top area of focus/investment and momentum in key businesses continues to build. While large markets like Japan, India, Germany, France and Canada have lagged in their adoption of Mako for various reasons, momentum in all is now building. For instance, in India, following significant implant price cuts in 2018, Stryker opted to deemphasize this market – but with economics here (both on the implant and Mako system sides) having improved, management noted that this is now one of the highest growth geographies for the recon franchise.
- Launch of the handheld Mako RPS robotic system continues to be very well received and which management expects to help sustain the share gain momentum in knee implants. Indeed, the company noted that a majority of early unit placements have been at competitive accounts. Stryker has not seen, nor expects to see, any cannibalization of Mako by the RPS launch.
- Rollout of Mako 4 continues to gain momentum, with management remaining bullish on the shoulder recon opportunity where expectations remain for Mako to yield market expansion given the highly challenging nature of the conventional approach that has limited the number of orthopedic surgeons doing the procedure. Mako should drive higher treatment rates.
- On soft tissue surgery, this remains an area of high interest and potential M&A opportunities continue to be evaluated, with management again noting that entry into this space would surely be via acquisition. He pointed out that, unlike J&J and Medtronic whose entry into this category was largely defensive given their strong incumbent presence in laparoscopic devices, such a move by Stryker would be purely offensive and hence will continue to be disciplined on this front.
- When asked about remanufactured da Vinci instruments, management once again confirmed that it is watching this space with high interest – and any such move here by Stryker would obviously have big implications for Intuitive.
- In terms of other robotic opportunities, management noted that robotic stroke intervention technologies are of high interest – and here too any such move here would be via acquisition vs organic R&D investment. Broadly speaking, stroke continues to be an increasing area of focus at SRS this year by other development-stage companies.

Medtronic Hugo

- While disclosures around Hugo adoption (installed base etc) continue to be frustratingly limited, directional commentary from management continues to be positive – particularly internationally where the system has now been approved/launched for some time and has broader clearances.
- While US approval is still limited to urology, clearances for general surgery and gynecology are in pursuit and once in hand will allow Hugo to address ~80% of the soft tissue robotic surgery market – and thus compel more hospitals/ASCs to evaluate the Medtronic platform.
- Management spun today's approval of J&J's Ottava as a positive for the overall robotic surgery market, arguing that technology penetration remains much more important than market share – also noting that global TAM penetration is still just single digits.

- Medtronic's marketing pitch centers around five main differentiating attributes of Hugo: (1) modularity (enables flexibility in the OR relative to port placement etc.); (2) open visualization console that better facilitates collaboration in the OR vs the immersive da Vinci console; (3) advanced instrumentation like Ligasure that has been used in over 35 million procedures; (4) digital surgery (Touch Surgery ecosystem now in over 1500 ORs); (5) partnership model with customers.

Appendix 1

Important Disclosures

*Other information available upon request

Disclosure checklist

Company	Ticker	Recent price*	Disclosure
Intuitive Surgical	ISRG.OQ	340.69 (USD) 22 Jul 2026	2, 14, 24

*Prices are current as of the end of the previous trading session unless otherwise indicated and are sourced from local exchanges via Reuters, Bloomberg and other vendors . Other information is sourced from Deutsche Bank, subject companies, and other sources. For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at https://research.db.com/Research/Disclosures/EquityResearchDisclosures. Aside from within this report, important risk and conflict disclosures can also be found at https://research.db.com/Research/Disclosures/Disclaimer. Investors are strongly encouraged to review this information before investing.

Important Disclosures Required by U.S. Regulators

Disclosures marked with an asterisk may also be required by at least one jurisdiction in addition to the United States.See Important Disclosures Required by Non-US Regulators and Explanatory Notes.

2. Deutsche Bank and/or its affiliate(s) may act as a market maker or liquidity provider in the financial instruments issued by this company.

14. Deutsche Bank and/or its affiliate(s) has received compensation from this company within the past year for non-investment banking related services.

Important Disclosures Required by Non-U.S. Regulators

Disclosures marked with an asterisk may also be required by at least one jurisdiction in addition to the United States.See Important Disclosures Required by Non-US Regulators and Explanatory Notes.

2. Deutsche Bank and/or its affiliate(s) may act as a market maker or liquidity provider in the financial instruments issued by this company.

24. Deutsche Bank and/or its affiliate(s) is or has been over the previous 12 months party to an agreement with the company relating to the provision of services set out in Sections A and B of Annex I of Directive 2014/65/EU, or has over the previous 12 months been obliged or entitled (as applicable) to pay or receive compensation relating to the provision of services set out in Sections A and B of Annex I of Directive 2014/65/EU.

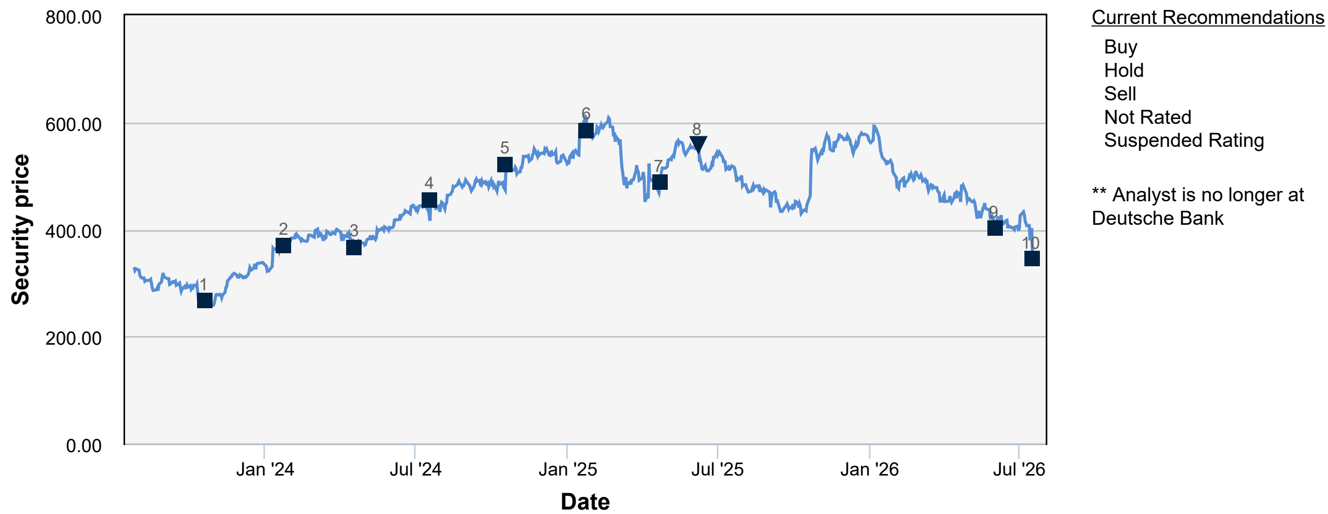
For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at https://research.db.com/Research/Disclosures/EquityResearchDisclosures. Aside from within this report, important risk and conflict disclosures can also be found at https://research.db.com/Research/Disclosures/Disclaimer. Investors are strongly encouraged to review this information before investing.

Analyst Certification

The views expressed in this report accurately reflect the personal views of the undersigned lead analyst(s) about the subject issuer and the securities of the issuer. In addition, the undersigned lead analyst(s) has not and will not receive any compensation for providing a specific recommendation or view in this report. Imron Zafar.

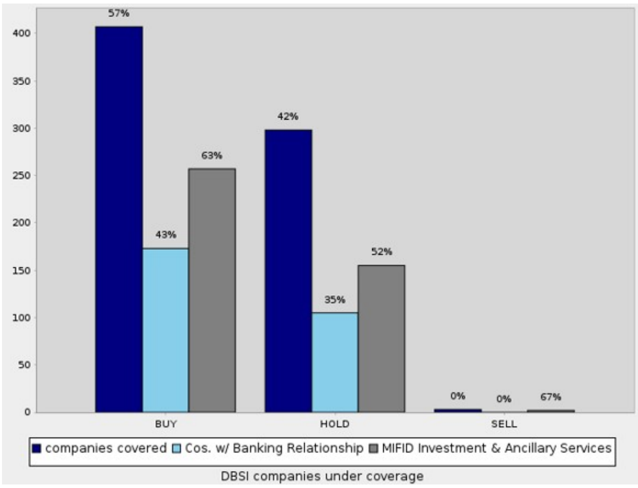
Historical recommendations and target price: Intuitive Surgical (ISRG.OQ)

(as of 07/22/2026)



1.	10/20/2023	Hold, Target Price Change USD 275.00, Current Price USD 266.91 Imron Zafar
2.	01/24/2024	Hold, Target Price Change USD 370.00, Current Price USD 370.07 Imron Zafar
3.	04/19/2024	Hold, Target Price Change USD 390.00, Current Price USD 366.34 Imron Zafar
4.	07/19/2024	Hold, Target Price Change USD 445.00, Current Price USD 455.01 Imron Zafar
5.	10/18/2024	Hold, Target Price Change USD 505.00, Current Price USD 521.15 Imron Zafar
6.	01/24/2025	Hold, Target Price Change USD 585.00, Current Price USD 584.05 Imron Zafar
7.	04/23/2025	Hold, Target Price Change USD 515.00, Current Price USD 487.93 Imron Zafar
8.	06/08/2025	Downgraded to Sell, Target Price Change USD 440.00, Current Price USD 557.08 Imron Zafar
9.	06/02/2026	Sell, Target Price Change USD 366.00, Current Price USD 402.30 Imron Zafar
10.	07/17/2026	Sell, Target Price Change USD 324.00, Current Price USD 345.42 Imron Zafar

Equity rating dispersion and banking relationships



Equity Rating and Dispersion Key

The Equity Rating Dispersion Chart depicts the following:

The proportion of recommendations that are rated "buy", "sell" and "hold" over the previous 12 months. This is shown for securities issued in the stated region e.g. "Europe Universe". See rating definitions below. This is represented by the "Companies Covered" bars in the chart. The percentage value displayed above the bar is the proportion as a percentage. E.g. 50% above the “buy” / “Companies Covered” bar means that 50% of DB’s equity research covered companies over the past 12 months have a “buy” rating.

Next to each of the three respective bars showing the proportion of "buy", "sell" and "hold" recommendations we provide two additional bars to show:

- The proportion of "buy", "sell" or "hold recommendations where Deutsche Bank and or/Affiliates provided MIFID Investment or Ancillary Services in the past 12 months. This is represented in the "MIFID Investment and Ancillary Services" bar. The percentage value displayed above the bar shows the proportion of Companies Covered with the given rating where DB has also provided MIFID Investment and Ancillary Services in the past 12 months. E.g. 50% above the “Cos. w/ MIFID Investment and Ancillary Services” bar means 50% of the Companies Covered with the rating stated have also received MIFID Investment and Ancillary Services from DB.

- The proportion of "buy" (or "sell" or "hold) recommendations where Deutsche Bank and or/Affiliates has provided Investment Banking services in the past 12 months for which it has received compensation. The percentage value displayed above the bar shows the proportion of Companies Covered with the stated rating where DB has also provided Investment Banking services in the past 12 months. E.g. 50% above the “Cos. w/ Investment Banking relationship” bar means 50% of the Companies Covered with the rating stated also have an Investment Banking Relationship with DB.

Buy: Based on a current 12- month view of TSR, we recommend that investors buy the stock.

Sell: Based on a current 12-month view of TSR, we recommend that investors sell the stock.

Hold: We take a neutral view on the stock 12-months out and, based on this time horizon, do not recommend either a Buy or Sell.

TSR = Total Shareholder Return. Percentage change in share price from current price to projected target price plus projected dividend yield

Newly issued research recommendations and target prices supersede previously published research.

Additional Information

The information and opinions in this report were prepared by Deutsche Bank AG or one of its affiliates (collectively 'Deutsche Bank'). Though the information herein is believed to be reliable and has been obtained from public sources believed to be reliable, Deutsche Bank makes no representation as to its accuracy or completeness. Hyperlinks to third-party websites in this report are provided for reader convenience only. Deutsche Bank neither endorses the content nor is responsible for the accuracy or security controls of those websites.

If you use the services of Deutsche Bank in connection with a purchase or sale of a security that is discussed in this report, or is included or discussed in another communication (oral or written) from a Deutsche Bank analyst, Deutsche Bank may act as principal for its own account or as agent for another person.

Deutsche Bank may consider this report in deciding to trade as principal. It may also engage in transactions, for its own account or with customers, in a manner inconsistent with the views taken in this research report. Others within Deutsche Bank, including strategists, sales staff and other analysts, may take views that are inconsistent with those taken in this research report. Deutsche Bank issues a variety of research products, including fundamental analysis, equity-linked analysis, quantitative analysis and trade ideas. Recommendations contained in one type of communication may differ from recommendations contained in others, whether as a result of differing time horizons, methodologies, perspectives or otherwise. Deutsche Bank and/or its affiliates may also be holding debt or equity securities of the issuers it writes on. Analysts are paid in part based on the profitability of Deutsche Bank AG and its affiliates, which includes investment banking, trading and principal trading revenues.

Opinions, estimates and projections constitute the current judgment of the author as of the date of this report. They do not necessarily reflect the opinions of Deutsche Bank and are subject to change without notice. Deutsche Bank provides liquidity for buyers and sellers of securities issued by the companies it covers. Deutsche Bank research analysts sometimes have shorter-term trade ideas that may be inconsistent with Deutsche Bank's existing longer-term ratings. Some trade ideas for equities are listed as Catalyst Calls on the Research Website (https://research.db.com/Research/) , and can be found on the general coverage list and also on the covered company's page. A Catalyst Call represents a high-conviction belief by an analyst that a stock will outperform or underperform the market and/or a specified sector over a time frame of no less than two weeks and no more than three months. In addition to Catalyst Calls, analysts may occasionally discuss with our clients, and with Deutsche Bank salespersons and traders, trading strategies or ideas that reference catalysts or events that may have a near-term or medium-term impact on the market price of the securities discussed in this report, which impact may be directionally counter to the analysts' current 12-month view of total return or investment return as described herein. Deutsche Bank has no obligation to update, modify or amend this report or to otherwise notify a recipient thereof if an opinion, forecast or estimate changes or becomes inaccurate. Coverage and the frequency of changes in market conditions and in both general and company-specific economic prospects make it difficult to update research at defined intervals. Updates are at the sole discretion of the coverage analyst or of the Research Department Management, and the majority of reports are published at irregular intervals. This report is provided for informational purposes only and does not take into account the particular investment objectives, financial situations, or needs of individual clients. It is not an offer or a solicitation of an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Target prices are inherently imprecise and a product of the analyst's judgment. The financial instruments discussed in this report may not be suitable for all investors, and investors must make their own informed investment decisions. Prices and availability of financial instruments are subject to change without notice, and investment transactions can lead to losses as a result of price fluctuations and other factors. If a financial instrument is denominated in a currency other than an investor's currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Performance calculations exclude transaction costs, unless otherwise indicated. Unless otherwise indicated, prices are current as of the end of the previous trading session and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Data is also sourced from Deutsche Bank, subject companies, and other parties. Artificial intelligence tools may be used in the preparation of this material, including but not limited to assist in fact-finding, data analysis, pattern recognition, content drafting and editorial corrections pertaining to research material.

The Deutsche Bank Research Department is independent of other business divisions of the Bank. Details regarding our organizational arrangements and information barriers we have to prevent and avoid conflicts of interest with respect to our research are available on our website (https://research.db.com/Research/) under Disclaimer.

Macroeconomic fluctuations often account for most of the risks associated with exposures to instruments that promise to pay fixed or variable interest rates. For an investor who is long fixed-rate instruments (thus receiving these cash flows), increases in interest rates naturally lift the discount factors applied to the expected cash flows and thus cause a loss. The longer the maturity of a certain cash flow and the higher the move in the discount factor, the higher will be the loss. Upside surprises in inflation, fiscal funding needs, and FX depreciation rates are among the most common adverse macroeconomic shocks to receivers. But counterparty exposure, issuer creditworthiness, client segmentation, regulation (including changes in assets holding limits for different types of investors), changes in tax policies, currency convertibility (which may constrain currency conversion, repatriation of profits and/or liquidation of positions), and settlement issues related to local clearing houses are also important risk factors. The sensitivity of fixed-income instruments to macroeconomic shocks may be mitigated by indexing the contracted cash flows to inflation, to FX depreciation, or to specified interest rates - these are common in emerging markets. The index fixings may - by construction - lag or mis-measure the actual move in the underlying variables they are intended to track. The choice of the proper fixing (or metric) is particularly important in swaps markets, where floating coupon rates (i.e., coupons indexed to a typically short-dated interest rate reference index) are exchanged for fixed coupons. Funding in a currency that differs from the currency in which coupons are denominated carries FX risk. Options on swaps (swaptions) the risks typical to options in addition to the risks related to rates movements.

Derivative transactions involve numerous risks including market, counterparty default and illiquidity risk. The appropriateness of these products for use by investors depends on the investors' own circumstances, including their tax position, their regulatory environment and the nature of their other assets and liabilities; as such, investors should take expert legal and financial advice before entering into any transaction similar to or inspired by the contents of this publication. The risk of loss in futures trading and options, foreign or domestic, can be substantial. As a result of the high degree of leverage obtainable in futures and options trading, losses may be incurred that are greater than the amount of funds initially deposited - up to theoretically unlimited losses. Trading in options involves risk and is not suitable for all investors. Prior to buying or selling an option, investors must review the 'Characteristics and Risks of Standardized Options", at https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document. If you are unable to access the website, please contact your Deutsche Bank representative for a copy of this important document.

Participants in foreign exchange transactions may incur risks arising from several factors, including the following: (i) exchange rates can be volatile and are subject to large fluctuations; (ii) the value of currencies may be affected by numerous market factors, including world and national economic, political and regulatory events, events in equity and debt markets and changes in interest rates; and (iii) currencies may be subject to devaluation or government-imposed exchange controls, which could affect the value of the currency. Investors in securities such as ADRs, whose values are affected by the currency of an underlying security, effectively assume currency risk.

Unless governing law provides otherwise, all transactions should be executed through the Deutsche Bank entity in the investor's home jurisdiction. Aside from within this report, important conflict disclosures can also be found at https://research.db.com/Research/ on each company's research page or under the 'Disclosures' tab. Investors are strongly encouraged to review this information before investing.

Deutsche Bank (which includes Deutsche Bank AG, its branches and affiliated companies) is not acting as a financial adviser, consultant or fiduciary to you or any of your agents (collectively, "You" or "Your") with respect to any information provided in this report. Deutsche Bank does not provide investment, legal, tax or accounting advice, Deutsche Bank is not acting as your impartial adviser, and does not express any opinion or recommendation whatsoever as to any strategies, products or any other information presented in the materials. Information contained herein is being provided solely on the basis that the recipient will make an independent assessment of the merits of any investment decision, and it does not constitute a recommendation of, or express an opinion on, any product or service or any trading strategy.

The information presented is general in nature and is not directed to retirement accounts or any specific person or account type, and is therefore provided to You on the express basis that it is not advice, and You may not rely upon it in making Your decision. The information we provide is being directed only to persons we believe to be financially sophisticated, who are capable of evaluating investment risks independently, both in general and with regard to particular transactions and investment strategies, and who understand that Deutsche Bank has financial interests in the offering of its products and services. If this is not the case, or if You are an IRA or other retail investor receiving this directly from us, we ask that you inform us immediately.

In July 2018, Deutsche Bank revised its rating system for short term ideas whereby the branding has been changed to Catalyst Calls ("CC") from SOLAR ideas; the rating categories for Catalyst Calls originated in the Americas region have been made consistent with the categories used by Analysts globally; and the effective time period for CCs has been reduced from a maximum of 180 days to 90 days.

United States: Approved and/or distributed by Deutsche Bank Securities Incorporated, a member of FINRA and SIPC. Analysts located outside of the United States are employed by non-US affiliates and are not registered/qualified as research analysts with FINRA.

European Economic Area (exc. United Kingdom): Approved and/or distributed by Deutsche Bank AG, a joint stock corporation with limited liability incorporated in the Federal Republic of Germany with its principal office in Frankfurt am Main. Deutsche Bank AG is authorized under German Banking Law and is subject to supervision by the European Central Bank and by BaFin, Germany's Federal Financial Supervisory Authority.

United Kingdom: Approved and/or distributed by Deutsche Bank AG acting through its London Branch at 21 Moorfields, London EC2Y 9DB. Deutsche Bank AG in the United Kingdom is authorised by the Prudential Regulation Authority and is subject to limited regulation by the Prudential Regulation Authority and Financial Conduct Authority. Details about the extent of our authorisation and regulation are available on request.

Hong Kong SAR: Distributed by Deutsche Bank AG, Hong Kong Branch except for any research content relating to futures contracts within the meaning of the Hong Kong Securities and Futures Ordinance Cap. 571. Research reports on such futures contracts are not intended for access by persons who are located, incorporated, constituted or resident in Hong Kong. The author(s) of a research report , and the entities for which they act and/or are affiliated with, may not be licensed to carry on regulated activities in Hong Kong and, if not licensed, do not hold themselves out as being able to do so. The provisions set out above in the 'Additional Information' section shall apply to the fullest extent permissible by local laws and regulations, including without limitation the Code of Conduct for Persons Licensed or Registered with the Securities and Futures Commission. This report is intended for distribution only to 'professional investors' as defined in Part 1 of Schedule of the SFO. This document must not be acted or relied on by persons who are not professional investors. Any investment or investment activity to which this document relates is only available to professional investors and will be engaged only with professional investors.

India: Prepared by Deutsche Equities India Private Limited (DEIPL) having CIN: U65990MH2002PTC137431 and registered office at 14th Floor, The Capital, C-70, G Block, Bandra Kurla Complex, Mumbai (India) 400051. Tel: + 91 22 7180 4444. It is registered by the Securities and Exchange Board of India (SEBI) as a Stock broker bearing registration no.: INZ000252437; Merchant Banker bearing SEBI Registration no.: INM000010833 and Research Analyst bearing SEBI Registration no.: INH000001741. DEIPL's Compliance / Grievance officer is Ms. Rashmi Poddar (Tel: +91 22 7180 4929 email ID: complaints.deipl@db.com). Registration granted by SEBI and certification from NISM in no way guarantee performance of DEIPL or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. DEIPL may have received administrative warnings from the SEBI for breaches of Indian regulations. Deutsche Bank and/or its affiliate(s) may have debt holdings or positions in the subject company. With regard to information on associates, please refer to the "Shareholdings" section in the Annual Report at: https://www.db.com/ir/en/annual-reports.htm. For latest India research audit report, refer https://country.db.com/india/deutsche-equities-india/index?language_id=1.

Japan: Approved and/or distributed by Deutsche Securities Inc.(DSI). Registration number - Registered as a financial instruments dealer by the Head of the Kanto Local Finance Bureau (Kinsho) No. 117. Member of associations: JSDA, Type II Financial Instruments Firms Association and The Financial Futures Association of Japan. Commissions and risks involved in stock transactions - for stock transactions, we charge stock commissions and consumption tax by multiplying the transaction amount by the commission rate agreed with each customer. Stock transactions can lead to losses as a result of share price fluctuations and other factors. Transactions in foreign stocks can lead to additional losses stemming from foreign exchange fluctuations. We may also charge commissions and fees for certain categories of investment advice, products and services. Recommended investment strategies, products and services carry the risk of losses to principal and other losses as a result of changes in market and/or economic trends, and/or fluctuations in market value. Before deciding on the purchase of financial products and/or services, customers should carefully read the relevant disclosures, prospectuses and other documentation. 'Moody's', 'Standard Poor's', and 'Fitch' mentioned in this report are not registered credit rating agencies in Japan unless Japan or 'Nippon' is specifically designated in the name of the entity. Reports on Japanese listed companies not written by analysts of DSI are written by Deutsche Bank Group's analysts with the coverage companies specified by DSI. Some of the foreign securities stated on this report are not disclosed according to the Financial Instruments and Exchange Law of Japan. Target prices set by Deutsche Bank's equity analysts are based on a 12-month forecast period.

Korea: Distributed by Deutsche Securities Korea Co.

South Africa: Deutsche Bank AG Johannesburg is incorporated in the Federal Republic of Germany (Branch Register Number in South Africa: 1998/003298/10).

Singapore: This report is issued by Deutsche Bank AG, Singapore Branch (One Raffles Quay #18-00 South Tower Singapore 048583, 65 6423 8001), which may be contacted in respect of any matters arising from, or in connection with, this report. Where this report is issued or promulgated by Deutsche Bank in Singapore to a person who is not an accredited investor, expert investor or institutional investor (as defined in the applicable Singapore laws and regulations), they accept legal responsibility to such person for its contents.

Taiwan: Information on securities/investments that trade in Taiwan is for your reference only. Readers should independently evaluate investment risks and are solely responsible for their investment decisions. Deutsche Bank research may not be distributed to the Taiwan public media or quoted or used by the Taiwan public media without written consent. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation to trade in such securities/instruments.

Qatar: Deutsche Bank AG in the Qatar Financial Centre (registered no. 00032) is regulated by the Qatar Financial Centre Regulatory Authority. Deutsche Bank AG - QFC Branch may undertake only the financial services activities that fall within the scope of its existing QFCRA license. Its principal place of business in the QFC: Qatar Financial Centre, Tower, West Bay, Level 5, PO Box 14928, Doha, Qatar. This information has been distributed by Deutsche Bank AG. Related financial products or services are only available only to Business Customers, as defined by the Qatar Financial Centre Regulatory Authority.

Russia: The information, interpretation and opinions submitted herein are not in the context of, and do not constitute, any appraisal or evaluation activity requiring a license in the Russian Federation.

Kingdom of Saudi Arabia: Deutsche Securities Saudi Arabia (DSSA) is a closed joint stock company authorized by the Capital Market Authority of the Kingdom of Saudi Arabia with a license number (No. 37-07073) to conduct the following business activities: Dealing, Arranging, Advising, and Custody activities. DSSA registered office is Faisaliah Tower, 17th Floor, King Fahad Road - Al Olaya District Riyadh, Kingdom of Saudi Arabia P.O. Box 301806.

United Arab Emirates: Deutsche Bank AG in the Dubai International Financial Centre (registered no. 00045) is regulated by the Dubai Financial Services Authority. Deutsche Bank AG - DIFC Branch may only undertake the financial services activities that fall within the scope of its existing DFSA license. Principal place of business in the DIFC: Dubai International Financial Centre, The Gate Village, Building 5, PO Box 504902, Dubai, U.A.E. This information has been distributed by Deutsche Bank AG. Related financial products or services are available only to Professional Clients, as defined by the Dubai Financial Services Authority.

Australia and New Zealand: This research is intended only for 'wholesale clients' within the meaning of the Australian Corporations Act and New Zealand Financial Advisors Act, respectively. Please refer to Australia-specific research disclosures and related information at https://www.dbresearch.com/PROD/RPS_EN-PROD/PROD0000000000521304.xhtml . Where research refers to any particular financial product recipients of the research should consider any product disclosure statement, prospectus or other applicable disclosure document before making any decision about whether to acquire the product. In preparing this report, the primary analyst or an individual who assisted in the preparation of this report has likely been in contact with the company that is the subject of this research for confirmation/clarification of data, facts, statements, permission to use company-sourced material in the report, and/or site-visit attendance. Without prior approval from Research Management, analysts may not accept from current or potential Banking clients the costs of travel, accommodations, or other expenses incurred by analysts attending site visits, conferences, social events, and the like. Similarly, without prior approval from Research Management and Anti-Bribery and Corruption ("ABC") team, analysts may not accept perks or other items of value for their personal use from issuers they cover.

Additional information relative to securities, other financial products or issuers discussed in this report is available upon request. This report may not be reproduced, distributed or published without Deutsche Bank's prior written consent.

Backtested, hypothetical or simulated performance results have inherent limitations. Unlike an actual performance record based on trading actual client portfolios, simulated results are achieved by means of the retroactive application of a backtested model itself designed with the benefit of hindsight. Taking into account historical events the backtesting of performance also differs from actual account performance because an actual investment strategy may be adjusted any time, for any reason, including a response to material, economic or market factors. The backtested performance includes hypothetical results that do not reflect the reinvestment of dividends and other earnings or the deduction of advisory fees, brokerage or other commissions, and any other expenses that a client would have paid or actually paid. No representation is made that any trading strategy or account will or is likely to achieve profits or losses similar to those shown. Alternative modeling techniques or assumptions might produce significantly different results and prove to be more appropriate. Past hypothetical backtest results are neither an indicator nor guarantee of future returns. Actual results will vary, perhaps materially, from the analysis.

The method for computing individual E,S,G and composite ESG scores set forth herein is a novel method developed by the Research department within Deutsche Bank AG, computed using a systematic approach without human intervention. Different data providers, market sectors and geographies approach ESG analysis and incorporate the findings in a variety of ways. As such, the ESG scores referred to herein may differ from equivalent ratings developed and implemented by other ESG data providers in the market and may also differ from equivalent ratings developed and implemented by other divisions within the Deutsche Bank Group. Such ESG scores also differ from other ratings and rankings that have historically been applied in research reports published by Deutsche Bank AG. Further, such ESG scores do not represent a formal or official view of Deutsche Bank AG. It should be noted that the decision to incorporate ESG factors into any investment strategy may inhibit the ability to participate in certain investment opportunities that otherwise would be consistent with your investment objective and other principal investment strategies. The returns on a portfolio consisting primarily of sustainable investments may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered, and the investment opportunities available to such portfolios may differ. Companies may not necessarily meet high performance standards on all aspects of ESG or sustainable investing issues; there is also no guarantee that any company will meet expectations in connection with corporate responsibility, sustainability, and/or impact performance.

David Folkerts-Landau
Group Chief Economist and Global Head of Research

Pam Finelli COO and Head of Fixed Income Research	Steve Pollard Global Head of Company Research and Sales	Jim Reid Global Head of Macro and Thematic Research	Tim Rokossa Head of European Company Research
Matthew Barnard Head of Americas Company Research	Debbie Jones Global Head of Sustainability and Data Innovation, Research	Robin Winkler Head of German Macro Research	Sameer Goel Global Head of EM & APAC Research
Francis Yared Global Head of Rates Research	George Saravelos Global Head of FX Research	Peter Hooper Vice-Chair of Research	Nilendra de-Mel Head of APAC & Middle East Product Development

International Production Locations

Deutsche Bank AG Deutsche Bank Place Level 16 Corner of Hunter & Phillip Streets Sydney, NSW 2000 Australia Tel: (61) 2 8258 1234	Deutsche Bank AG Equity Research Mainzer Landstrasse 11-17 60329 Frankfurt am Main Germany Tel: (49) 69 910 00	Deutsche Bank AG Filiale Hongkong International Commerce Centre, 1 Austin Road West,Kowloon, Hong Kong Tel: (852) 2203 8888	Deutsche Securities Inc. 1-3-1 Azabudai Azabudai Hills Mori JP Tower Minato-ku, Tokyo 106-0041 Japan Tel: (81) 3 6730 1000
Deutsche Bank AG 21 Moorfields London EC2Y 9DB United Kingdom Tel: (44) 20 7545 8000	Deutsche Bank Securities Inc. The Deutsche Bank Center 1 Columbus Circle New York, NY 10019 Tel: (1) 212 250 2500	Deutsche Bank AG Filiale Singapur One Raffles Quay, South Tower, Singapore 048583 TeL: +65 6423 8001	